$$\int_{-1}^{1} x^2 f(x)\,dx = A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2) + R(f)$$

$m=2$ 3 noduri x_0, x_1, x_2 $w(x)=x^2$ funcția pondere

3 coeficienți A_0, A_1, A_2

GRADUL MAXIM $= 2m+1 = 2\cdot 2+1 = 5$

Soluția I ("de avarie") NECUNOSCUTE $x_0, x_1, x_2, A_0, A_1, A_2$

6 necunoscute $\Rightarrow$ avem nevoie de 6 ecuații

$$\begin{cases} R(x^0)=0 \\ R(x^1)=0 \\ R(x^2)=0 \\ R(x^3)=0 \\ R(x^4)=0 \\ R(x^5)=0 \end{cases}$$

$f=x^0=1$ $2\int_0^1 x^2 dx = \int_{-1}^{1} x^2\cdot 1\,dx = A_0\cdot 1 + A_1\cdot 1 + A_2\cdot 1$

$f=x$ $0=\int_{-1}^{1} x^2\cdot x\,dx = A_0 x_0 + A_1 x_1 + A_2 x_2$

$f=x^2$ $2\int_0^1 x^4 dx = \int_{-1}^{1} x^2\cdot x^2\,dx = A_0 x_0^2 + A_1 x_1^2 + A_2 x_2^2$

$f=x^3$ $0=\int_{-1}^{1} x^2\cdot x^3\,dx = A_0 x_0^3 + A_1 x_1^3 + A_2 x_2^3$

$f=x^4$ $\int_{-1}^{1} x^2\cdot x^4\,dx = A_0 x_0^4 + A_1 x_1^4 + A_2 x_2^4$

$f=x^5$ $0=\int_{-1}^{1} x^2\cdot x^5\,dx = A_0 x_0^5 + A_1 x_1^5 + A_2 x_2^5$

EX 2) $$\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} f(x)\,dx = \underset{A_0}{A} f(\underset{x_0}{-1}) + \underset{A_1}{B} f(\underset{x_1}{\alpha}) + \underset{A_2}{C} f(\underset{x_2}{1}) + R(f)$$

A, B, C, α – 4 necunoscute a.î. gradul să fie maxim

$w(x) = \frac{1}{\sqrt{1-x^2}}$ | Pt. $n=2$ gradul maxim posibil ar fi fost $5 = 2\cdot 2+1$ |

$n=2$ Soluția I ("de avarie")

4 necunoscute $\Rightarrow$ 4 ecuații $\Rightarrow$ $\begin{cases} R(x^0)=0 \\ R(x^1)=0 \\ R(x^2)=0 \\ R(x^3)=0 \end{cases}$ $\Rightarrow$ gradul maxim care poate fi obținut este 3

$f=1=x^0$ $\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\cdot 1\,dx = A\cdot 1 + B\cdot 1 + C\cdot 1$

$f=x^1$ $0=\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} x\,dx = A(-1) + B\cdot\alpha + C\cdot 1$

$f=x^2$ $\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} x^2\,dx = A\cdot(-1)^2 + B\cdot\alpha^2 + C\cdot 1^2$

$f=x^3$ $0=\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} x^3\,dx = A(-1)^3 + B\cdot\alpha^3 + C\cdot 1^3$

P.D.f. "Mai elegant"

$R(1)=R(x)=R(x^2)=R(x^3)=0 \iff R(P)=0 \ \forall P\in\Pi_3$

$f=(x-1)(x+1) \rightarrow f(-1)=0,\ f(1)=0$

$$\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}(x-1)(x+1)\,dx = A\cdot 0 + B(\alpha-1)(\alpha+1) + C\cdot 0$$

$$\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}(x-1)(x+1)\,dx = B(\alpha^2-1)$$

"și mai elegant" $f(x)=(x-1)(x+1)(x-\alpha)\in\Pi_3$

$f(-1)=f(1)=f(\alpha)=0$

$$\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}(x-1)(x+1)(x-\alpha)\,dx = A\cdot 0 + B\cdot 0 + C\cdot 0$$

$$\int_{-1}^{1} \frac{(x-1)(x+1)}{\sqrt{1-x^2}} x\,dx = \alpha\int_{-1}^{1} \frac{(x-1)(x+1)}{\sqrt{1-x^2}}\,dx \Rightarrow \alpha = \ldots$$

$f=1, f=x, f=x^2 \Rightarrow A, B, C,$

Revenim la ex 1 și ex 2:

$$\int_{-1}^{1} x^2 f(x)\,dx = A_0 f(x_0) + A_1 f(x_1) + A_2 f(x_2) + R(f)$$

$$\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}} f(x)\,dx = A_0 f(-1) + A_1 f(x_1) + A_2 f(1) + R(f)$$

"Poza 2" TEORIE

$$\int_a^b w(x) f(x) = \sum_{i=0}^{m} A_{m,i} f(a_i) + \sum_{j=1}^{k} B_{k,j} f(b_j) + R(f)$$

$a_0, a_1, \ldots, a_m,\ b_1, b_2, \ldots, b_k \rightarrow m+1+k$ noduri

$A_{m,0}, A_{m,1}, \ldots A_{m,m}\ B_{k,1}, B_{k,2} \ldots B_{k,k} \rightarrow m+1+k$ coeficienți

$n = m+k$

GRADUL MAXIM POSIBIL $2n+1 = 2\cdot m + 2\cdot k + 1$

$b_1, b_2, \ldots, b_k$ – se presupun cunoscute $\Rightarrow$ gradul maxim va fi $= 2m+k+1$ (nr. necunoscute -1)

EX 1: $m=2$ $k=0$ $w(x)=x^2$

$\underbrace{x_0, x_1, x_2}_{a_0, a_1, a_2}$ $\underbrace{A_0, A_1, A_2}_{A_{m,0}, A_{m,1}, A_{m,2}}$

Ex 2: $$\int_{-1}^{1} \underbrace{\frac{1}{\sqrt{1-x^2}}}_{w(x)=} f(x)\,dx = B\cdot f(\alpha) + \left(A f(-1) + C f(1)\right) + R(f)$$

$m=0$ $\underbrace{\alpha}_{a_0}$ $\underbrace{B}_{A_{m,0}}$ $k=2$ $\underbrace{-1, 1}_{b_1, b_2}$ $\underbrace{A \quad C}_{B_{k,1}\ B_{k,2}}$

g. max $= 2\cdot 0 + 2+1 = 3$

TEORIE

$a_0, a_1, \ldots, a_m$ ($m+1$ noduri necunoscute) vor fi rădăcinile polinomului ortogonal de grad $m+1$ relativ la ponderea

$$\overline{w}(x) = w(x)(x-b_1)(x-b_2)\ldots(x-b_k)$$

$P_0, P_1, \ldots, P_{m+1}, \ldots$

$$\int_a^b \overline{w}(x) P_i(x)\cdot P_j(x)\,dx = 0, \text{ pt } i\neq j$$

EX 1 $w(x)=x^2$, $k=0$

x_0, x_1, x_2 ($m=2$) sunt rădăcinile polinomului de grad 3 relativ la ponderea $\overline{w}(x)=x^2$

Ex 2 $w(x)=\frac{1}{\sqrt{1-x^2}}$, $-1, 1$

$x_0=\alpha$ ($m=0$), $k=2$

α va fi rădăcina polinomului ortogonal de grad 1 relativ la ponderea

$$\overline{w}(x) = \frac{1}{\sqrt{1-x^2}}(x+1)(x-1)$$

Ex 3) $$\int_{-2}^{3} \underbrace{(x-2)e^x}_{w(x)=} f(x)\,dx = A_0 f(x_0) + A_1 f(x_1) + A_3 f(4) + A_4 f(5) + R(f)$$

$\underbrace{x_0, x_1}$ $4, 5$

$\searrow$ vor fi rădăcinile polinomului ortogonal de grad 2 relativ la ponderea $\overline{w}(x) = (x-2)e^x(x-4)(x-5)$

$$f(x) = \begin{cases} 5|2x+1| + 6|4x-3|, & x\in[0,1) \\ x^2+10, & x\in[1,2] \end{cases}$$

$S_{\Delta,1}(f) = S_{\Delta_4}(f)$ – splineul de ordin 1

$$\Delta_4 = \overset{x_0}{0} < \overset{x_1}{\tfrac{1}{2}} < \overset{x_2}{\tfrac{2}{3}} < \overset{x_3}{1} < \overset{x_4}{2}$$

$f(0)$ $f(\frac{1}{2})$ $f(\frac{2}{3})$ $f(1)$ $f(2)$

$a = x_0 < x_1 < \ldots < x_n = b$

$$S_{\Delta,1}(f)(x) = f(x_0)\ell_0(x) + f(x_1)\ell_1(x) + \ldots + f(x_n)\ell_n(x)$$

$\ell_0, \ell_1, \ldots, \ell_n$ – NU SUNT POLINOAMELE FUNDAMENTALE LAGRANGE

$$\int_0^2 S_{\Delta,1}'^2(x)\,dx = S_{\Delta,1} S_{\Delta,1}'\Big|_0^2 - \int_0^2 S_{\Delta,1} \underbrace{S_{\Delta,1}''}_{0}\,dx = S_{\Delta,1}^2(x)\Big|_0^2$$

$f = S_\Delta'$, $g' = S_\Delta'$